RITFORI – Let's Encrypt certficate software for the IBM i
Open Source – Created by Rowton IT Solutions Ltd and Forever-i Ltd

This has been tested on IBM i OS V7.4 and V7.5 but failed using V7.6 (to be looked at later).

Requirements
OS
- 5770SS1 option 3 - Extended Base Directory Support
- 5770SS1 option 12 - Host Servers
- 5770SS1 option 30 - Qshell
- 5770SS1 option 33 - PASE
- 5770SS1 option 34 - Digital Certificate Manager
- 5733SC1 *BASE - IBM Portable Utilities for i
- 5733SC1 option 1 - OpenSSH, OpenSSL, zlib
- 5770DG1 - IBM HTTP Server for IBM i
- 5770JV1 option 19 - IBM Java SE 11 64 bit

PTFs (14/03/2024)
7.5 SI85817
7.4 SI85818

IBM Open Source
ACS Open Source Package Management – check for the following
bash
curl
db2util
gawk
openssl
sed-gnu – installs as "sed" and "gsed" in /QOpenSys/pkgs/bin
unzip
wget

Bold are headings
Green text is a command or instruction to be executed or followed
Yellow text is an an important instruction.
Orange text has to be changed to your preferences.
Grey text are headings to aid clarity
Under Lined are commands to be entered into a 5250 screen

External software
- https://letsencrypt.org/ Issues and renews trusted certificates for 90 days for registered domains - Online
- https://github.com/acmesh-official/acme.sh (included) - Software to issue and renew certificates – Downloaded in Setup

Setup Instructions

If you downloaded and used this software before 6th Jan 2026, and want to upgrade, then follow instructions in the **Instructions-Copy_and_Paste.txt** document.

The Log files are in /HOMEDIR/ACME/LOG

You will need at least one registered domain name with a provider that has an API token for use with Let's Encrypt or other TLS certificate providers. Currently it only works for Cloudflare but it can be changed to add any of the 200 providers that are compatible with ACME. Please add an issue if you require one or two.

The domain name does not have to be used for public facing purposes. The domain name should be linked with the IBM i partition preferably by DNS.

The first domain name must be the one to be used for the RSE API
e.g. https://domain:2012/openapi/ui/
This is used for the automation of certificates in the DCM using the JKS. The JKS will be updated when this domain certificate is renewed.

Restart if you encounter problems

To restart from the beginning :-
Delete /HOMEDIR directory
Delete library RITFORI

Create User ID for Let's Encrypt

N.B. The homedir can be anything, if omitted it will default to /home/usrprf. The library will be RITFORI.
The user can be RITFORI, but it doesn't have to be. Do not use special characters except for _ in USRPRF. This user must be used for all programs, otherwise they will fail.

CRTUSRPRF USRPRF(USRPRF) PASSWORD(password) CURLIB(RITFORI) SPCAUT(*ALLOBJ *SECADM *AUDIT) HOMEDIR('/homedir')

The resulting messages are :-
Object RITFORI in library QSYS not found. +
User class and special authorities do not match system supplied values. +
User profile usrprf created.
they will be created shortly

Download the Repository Zip file

```
Download RITFORI code to your pc from
https://github.com/RitFori/RitFori/archive/refs/heads/main.zip
 (just put the link into a browser)

Upload RitFori-main.zip to /tmp on the IBM i
 (ACS – Integrated File System is good for this)
```

Unzip the Code to the IBM i into /tmp/ritfori

5250 Signon as your USRPRF

```
CD DIR('/tmp')
CALL PGM(QP2TERM)
PATH=/QopenSys/pkgs/bin:\$PATH
export PATH PASE_PATH
unzip ./RitFori-main.zip
mv RitFori-main ritfori
F3
```

Setup the USERID for bash, create IFS directories under /homedir and RITFORI library

5250 Still Signed on as your USRPRF

CRTBNDCL PGM(QGPL/RITFORISU) SRCSTMF('/tmp/ritfori/code/RitforiSU.clle') DBGVIEW(*SOURCE)

CALL PGM(QGPL/RITFORISU)

At this point there will be a log file in /homedir/setupUser.log and /homedir/acme/logacme_setup1.log. They should be empty if all has gone well, otherwise it will list any errors.

Setup the ACME software, register for an ACME account, create a new certificate and create a JKS for RSE API (HTTP Admin5)

The e-mail address is used to setup a Let's Encrypt account. They no longer send e-mails. You can create more than one account with the same e-mail, but you only need one account per partition.

It might help if you enter CALL QCMD first as it gives more space for long commands.
This may take a while (e.g. 3 to 4 minutes)

The jks_password can be anything you like.

The domain_token can be obtained from your domain name provider if they have an API service. In this case the token is enclosed in single quotes ' and double quotes ".

Double check you parameters.

ACMESETUP2 EMAIL('e-mail')
CERTNAME('name.domain.TLD')
TOKEN('"domain_token"')
JKSPASS('jks_password')

If you do not want to use DCM then you can stop here.
Otherwise setup RSE API certificate for executing DCM APIs as follows :-

DCM Setup part 1 – Manual – Register the JKS with the IBM RSE API (HTTP Admin5)

N.B. The Userprf does not have *IOSYSCFG so it cannot use HTTP Admin. Sign in with a different user or add *IOSYSCFG to you usrprf.

N.B. Some of the pages below change depending on whether you have previously set up a certificate for RSE API / Admin5, e.g. sections 6 and 7 don't appear. If you have problems it is best to DISABLE TLS, then restart Admin5. Close down any IBM i ACS connections and start again.

Use ACS →
Web (HTTP) Administration for i (which will bring up a browser with –)
https://IP or HOST:2010/HTTPAdmin or
http://IP or HOST:2001/HTTPAdmin →
Logon as any user with appropriate security (*ALLOBJ and *IOSYSCFG)
Manage →
Application Server →
Server = Admin 5 – V8.5 (int app svr) →

Disable TLS
Page = Disable TLS – *Welcome – Step 1 of 4*
Next →
If error at the bottom “There is no TLSconfiguration found for this application server”
GOTO Configure TLS – Welcome (below)
otherwise :-
Page = Disable TLS – *Disable TLS port – Step 2 of 4*
TICK 2012
Next →
Page = Disable TLS – *Restart the server now? – Step 3 of 4*
LEAVE Restart the server later by yourself
Next →
Page = Disable TLS – *Summary – Step 4 of 4*
Finish →
Configure TLS – Welcome
Page = Configure TLS – *Welcome – Step 1 of 10*
Next →
Page = Configure TLS – *Specify TLS port and protocol – Step 2 of 10*
TLS Port = 2012
TLS Product = TLSv1.3
LEAVE Yes, disable non-TLS port while configuring TLS port
Next →
Page = Configure TLS – *Specify keystore information – Step 3 of 10*
TICK Specify keystore path and type
Keystore path = /homedir/acme/data/certs/name.domain.TLD_ecc/lekeystore.jks
Keystore type = JKS
LEAVE BLANK Specify different path for truststore
Next →
Page = Configure TLS – *Specify keystore password– Step 4 of 10*
Password = jks_password
maybe – Confirm password = jks_password *Step 5 of 10*
Next →
Page = Configure TLS – *Specify Digital Certificate – Step 6 of 10*
TICK Select existing certificate from the keystore
Digital certificate = name.domain.TLD (select from drop down)
Next →
Page = Configure TLS – *Add trusted CA certificates – Step 7 of 10*
LEAVE No trust certificates to import
Next →
Page = Configure TLS – *Specify cipher for TLS – Step 8 of 10*
LEAVE Default ciphers
Next →
Page = Configure TLS – *Restart the server now? – Step 9 of 10*
TICK Restart the server immediately after the wizard (unless there is a good reason not to)
Next →
Page = Configure TLS – *Summary – Step 10 of 10*
Check the information
Finish →
Press refresh after 10 seconds or so Once it is started
Go to Browser to test the web page works
URL = https://name.domain.TLD:2012/openapi/ui/

DCM Setup part 2 – Manual – Setup an Application in the DCM (if not already there)

Use ACS →
Digital Certificate Manager (which will bring up a browser with –)
https://IP or HOST:2007/dcm/mainframe/system or
http://IP or HOST:2006/dcm/mainframe/system

Logon as any user with appropriate security (*ALLOBJ and *IOSYSCFG) as above
- Click Open Certificate Store
- Click *SYSTEM
- Enter the DCM *SYSTEM password (also needed in DCM_SETUP3)
- Click Manage Application Definitions
- Click Create
- Click Server
- Enter ID in UPPERCASE e.g. QIBM_HTTP_SERVER_YOURAPP or
  CERT_TEST_APP (to be entered in DCM_SETUP3 below)
- Enter Description → Specify → Description e.g. LE Test certificate application
- Click Create (at the bottom)
- Click Logout (top left)

DCM Setup part 3 – Install Let's Encrypt CAs and your Domain certificate with an Application

Sign onto a 5250 screen using the usrprf

It might help if you enter CALL QCMD and F11 first as it gives more space for long commands.
This may take a while (e.g. 3 to 4 minutes)

DCM_SETUP3 CERTNAME('name.domain.TLD')
USRPWD('usr_password')
DCMPWD('DCM_password')
CERTAPP('certificate_app_name')

All certificates will now be in the DCM and JKS.

Restart your application when you are ready to use the new certificate.

CALL PGM(LE_ADMIN5)
Create programs to restart your own appliactions

The following can be put into the Job Scheduler N.B. The standard job scheduler will do *MONTHLY

Renew the certificate before 90 days (30 or 60 days recommended)

LERenew CERTNAME('name.domain.TLD')
DCM_YN(YES)

Restart HTTP ADMIN5 in order to pick up the new certificate

CALL PGM(LE_ADMIN5)

Restart your application to use the new certificate

You will have to create your own CL code for this.

Delete old DCM certificate recommended before it expires

DCMDltCert CERTNAME('name.domain.TLD')